Unit 1

Lesson 2

Spark The Challenge!

Turning Ideas into Action.

Time Frame: [20 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Understand the problem and ask simple questions about how to make a playground better for everyone.
- Recognize the role of an accessibility specialist and how they help make playgrounds easier for all kids to use.
- Think of different ideas to make playgrounds fun and fair for everyone, including kids who may need extra help.
- Make a simple plan to design a playground and draw what they want to build.
- Talk about what worked well in the design and what can be improved to make the playground better for all kids.

Challenge questions of the lesson

1. What could make the playground easier to use for kids who can’t climb or run?
2. If you were a playground helper, what would you add to make sure all kids can play?
3. What part of your design helps kids who need extra help feel included and happy

SEL Relation

- Community Awareness (Understanding our world): We will learn about the people who help us in our community and why their jobs are important.
- Responsibility (Doing our part): We will think about how everyone, even us, has a responsibility to keep the playground safe and fun.
- Collaboration (Working together): By sharing what we learn, we can all get new ideas about how to make the playground better together.
- Appreciation (Feeling thankful): We can learn to appreciate the work that different community helpers do for us.

Engage

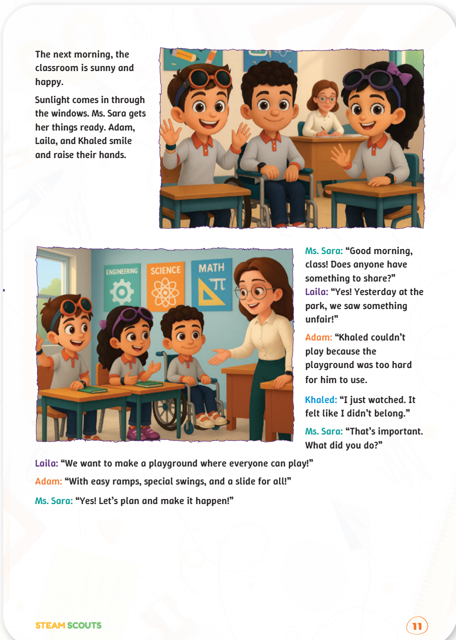
Teacher Role:

- **Storyteller:** Begin by reading or narrating a story that presents a real-world problem (e.g., a child can't access the playground). Use tone, visuals, and body language to engage students emotionally.
- **Bridge Builder:** Connect the problem in the story to the real world by saying,
- **Career Connector:** Introduce the concept of an accessibility specialist. Explain in simple terms:
- **Discussion Leader:** Ask reflection questions such as:
- **Context Builder:** Lead into the lesson activity by connecting the role to student work:

Expected From Students:

- **Active Listening:** Students will listen carefully to the story and pay attention to the problem faced by the characters.
- **Career Awareness:** Students will learn and explain what an accessibility specialist does in their own words.
- **Empathy and Connection:** Students will relate the specialist's role to their own challenge of designing an inclusive playground.
- **Idea Generation:** Students will begin to brainstorm features that would help all kids play safely and happily.
- **Participation:** Students will contribute to classroom discussion by sharing their ideas, asking questions, or reflecting on how they can help others through design.

Parts of the books included.



The next morning, the classroom is sunny and happy.

Sunlight comes in through the windows. Ms. Sara gets her things ready. Adam, Laila, and Khaled smile and raise their hands.

Ms. Sara: "Good morning, class! Does anyone have something to share?"
Laila: "Yes! Yesterday at the park, we saw something unfair!"

Adam: "Khaled couldn't play because the playground was too hard for him to use.

Khaled: "I just watched. It felt like I didn't belong."

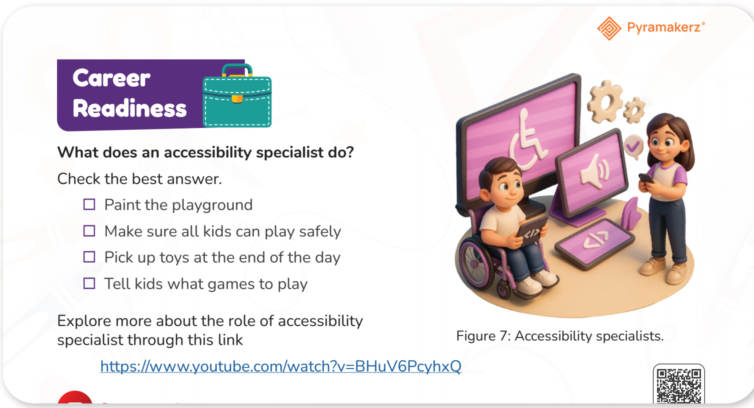
Ms. Sara: "That's important. What did you do?"

Laila: "We want to make a playground where everyone can play!"

Adam: "With easy ramps, special swings, and a slide for all!"

Ms. Sara: "Yes! Let's plan and make it happen!"

STEAM SCOUTS 11



Pyramakerz®

Career Readiness

What does an accessibility specialist do?

Check the best answer.

- ☐ Paint the playground
- ☐ Make sure all kids can play safely
- ☐ Pick up toys at the end of the day
- ☐ Tell kids what games to play

Explore more about the role of accessibility specialist through this link

https://www.youtube.com/watch?v=BHuV6PcyhxQ

Figure 7: Accessibility specialists.

Story

1. **Start with a Warm Welcome and Story Recap:**

"Good morning, engineers! Last sessiom, we started a story about our friends Adam, Laila, and Khaled at the playground. Do you remember what happened? Who remembers why Khaled felt sad?"

(Allow a few responses to activate memory and emotional connection.)

2. **Continue the Story with Expression:**

"Well, today is a brand-new day in their classroom. The sun is shining... and something exciting is about to happen. Let's see what they decided to do!"

(Read the story continuation aloud with dramatic tone and character voices.)

Possible Strategies for Implementation:

1. **Role-Play with Dialogue Cards**
   - Hand out simple line cards from the script and let students act out the scene.
   - Use props (e.g., pretend classroom, name tags) to immerse them in the story.
2. **Turn and Talk: "What Would You Do?"**
   - Prompt:
     "If you were in Miss Sarah's class, what would you build for Khaled?"
   - Let students discuss ideas with a partner for 2 minutes.

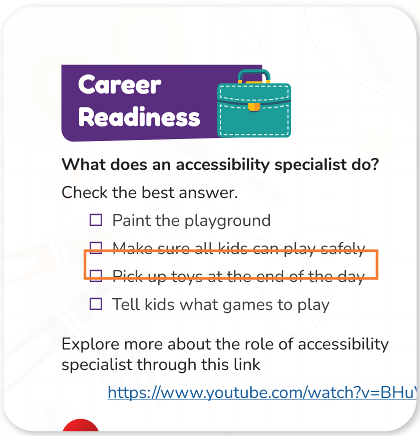
Career Readiness

Objective:

Students will understand the role of an accessibility specialist by identifying their main responsibility—ensuring that playgrounds and other spaces are safe and usable for all children.

This connects the lesson to a real-world career that focuses on inclusive design.

Answer:



Possible Strategies for Implementation:

1. **Think-Pair-Share with Reflective Reveal**

How to Implement:

- Begin by asking:
  "What do you think an accessibility specialist does?"

- Students first think quietly for 1 minute.
- Then, they pair up to discuss their ideas.
- Invite 2–3 pairs to share out with the whole class.

Enhancement:

- Before revealing the correct answer, tell students:

  "Let's watch a short video to see if our ideas were correct—or if we'll learn something new!"

Play the video: https://www.youtube.com/watch?v=BHuV6PcyhxQ

- After watching, ask:

  "Did your answer change? What did you learn from the video?"

  This strategy supports reflection, listening, and correcting misconceptions through student-centered learning.

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2. Role-Play with Real-World Connection

How to Implement:

- Set up a role-play:
  - One student is the "accessibility specialist."
  - The rest are "playground designers."
- The specialist gives advice such as:

"We need wide ramps here."

"Don't forget a quiet spot for kids who need calm."

"Can everyone reach the swing?

Enhancement:

- Allow the specialist to use notes or facts they remembered or learned from the video: https://www.youtube.com/watch?v=BHuV6PcyhxQ
- Rotate roles so more students can try giving expert advice.

Explain

Teacher Role:

- **Facilitator:** Introduce the video and guide students through the pre-, during-, and post-watching activities, connecting it to the EDP.
- **Purpose Setter:** Clearly explain the learning objectives and what students should be looking for in the video, emphasizing how the video demonstrates the EDP in action.
- **Engagement Encourager:** Prompt students to actively observe and think critically while watching, relating the video content to the EDP steps.
- **Discussion Moderator:** Lead the post-watching discussion, ensuring all students have an opportunity to share their observations and insights related to the EDP and playground design.
- **Concept Clarifier:** Address any questions or misunderstandings that may arise after watching the video, particularly about the EDP.
- **Connection Maker:** Help students connect the information in the video to previous activities and discussions about inclusivity and the EDP.

Expected From Students:

- **Active Viewing:** Students will watch the video attentively, focusing on identifying features that promote inclusive play and how the video demonstrates the EDP.
- **Note-Taking (Optional but encouraged):** Students may jot down key features, ideas, or questions that arise during the video, specifically noting how the EDP steps are being applied (even if not explicitly named).
- **Thoughtful Participation**: Students will contribute meaningfully to the pre- and post-watching discussions, sharing their initial thoughts, observations, and reflections.
- **Critical Analysis:** Students will think critically about the information presented in the video and how it relates to creating fair and accessible playgrounds, as well as how the EDP is used in the design process.

Parts of the books included.


Mastermind

Let's act like specific community helpers and think about how we can make the playground fun for everyone!

- Pick a role to learn about:

Engineers make sure the playground is safe and strong.

Doctors help keep kids safe and healthy.

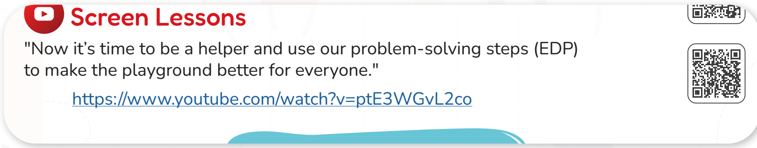
Policemen keep kids safe while they play.

Family helps kids have fun and take care of the playground.

Or any different community helper.

- Learn about your role at home.
- Share what you learned with the class next time.

Figure 8: Different community helpers — an artist, nurse, firefighter, soldier, police officer, and teacher.


Screen Lessons

"Now it's time to be a helper and use our problem-solving steps (EDP) to make the playground better for everyone."

https://www.youtube.com/watch?v=ptE3WGvL2co

Screen Lessons

Objective: Students will understand and apply the basic steps of the Engineering Design Process (EDP)—Ask, Imagine, Plan, Create, and Improve—by watching a kid-friendly video. Through this video, they will learn how to be helpers and problem-solvers who work to make playgrounds better for everyone, including those who may have different needs. The goal is to inspire students to think like young engineers and use simple design thinking to solve real-world challenges in inclusive ways.

Video link: https://youtu.be/ptE3WGvL2co?si=8mIzpYCsscb6W7br

Possible Strategies for implementation:

1. **Pre-Watching: Setting the Stage for the EDP.**
   - Strategy: Anticipation Guide (Modified for EDP Focus)
     - o How to Implement: Before showing the video, present students with statements related to the EDP and playground design. Examples:
       1. "When we design a playground, we don't need to think about all the steps."
       2. "Testing our playground designs is a waste of time."
       3. "It's important to ask questions and try different ideas when we are designing."
     - o Ask students to agree or disagree with each statement and briefly explain their reasoning before watching the video.
     - o Explain that the video will show how people use steps to make a playground better.
     - o Benefit: Introduces the idea of a process (the EDP) and connects it to the playground design task.
2. **During Watching: Active Engagement with the EDP.**
   - Strategy: Focused Observation with EDP Prompts
     - o How to Implement: Provide students with questions or prompts to focus on while watching, linking the video content to the EDP:
       1. "What problem are the people in the video trying to solve?" (Identifying the Need/Problem)
       2. "What are some of the ideas they come up with to solve the problem?" (Brainstorming Solutions)

3. "How do they decide which ideas are best?" (Selecting a Solution)
4. "Do they build or test anything in the video? What happens?" (Building and Testing)
5. "Do they make any changes? Why?" (Improving)

Write with them these notes on the board.

- Benefit: Helps students see the EDP in action and connect it to the task of designing an inclusive playground.

3. Post-Watching: Discussion and Reflection on the EDP

- Strategy 1: EDP Connection Chart
  - How to Implement: Open the EDP chart on the board or ask them to look in the book.
    1. Ask: What's the problem?
    2. Imagine: What are our ideas?
    3. Choose: Which idea is best?
  - Tell them that we will return back to the rest of the steps while we are making our playground.
  - Benefit: Reinforces the steps of the EDP and their application to playground design.

Engineering Design Process

Name: ....

Project: ....

Ask

How can we design a playground where all kids can play safely, including kids who use wheelchairs or have vision challenges?

Imagine

I can build spiral ramps so kids in wheelchairs can move up and down easily. I can add a merry-go-round with wide spaces so everyone can sit safely. I'll put a see-saw that can carry a wheelchair. I'll use a swing that lights up and makes sound for kids with vision problems. I'll make a sound path for blind children to follow?

Plan

Spiral ramps leading to slides and other play areas A wide merry-go-round in the center A see-saw with one side flat for a wheelchair Sound and light swing near the tree A path with musical stepping stones for blind kids.

Create

Improve

Mastermind

Objective:

Students will explore the roles of different community helpers and how each contributes to making playgrounds safe, inclusive, and fun for everyone. By choosing a role, researching it at home, and presenting it while dressed in costume, students will develop an understanding of civic responsibility, empathy, and collaboration. The activity also encourages oral communication, creativity, and personal connection to real-world helpers.

Activity Instructions

Step 1: Introduce the Activity (in class).

Say to students:

"Next class, you'll come back as a community helper! You'll choose a job—like a doctor, engineer, police officer, or even a family helper—and learn how they help make the playground better and safer for everyone."

- Show visuals or short video clips of different helpers in action.
- Give simple explanations (age-appropriate) of each role.

Step 2: Distribute the Role Sheet or Guide

- Hand out a worksheet or instruction card with:
  - o List of roles to choose from
  - o One question prompt:

"How does this helper make the playground fun and safe for everyone?"

  - o A space to draw or write one thing they learned

Step 3: Explain What to Do at Home

- **Say to students:**
- "At home, you'll pick one role and learn about it. Ask your family to help you find something out—like reading a book, watching a video, or talking together."
- "Then, come to school dressed like your helper! You can bring a costume, a tool they use, or even just a drawing of them."

Step 4: Share Presentation Expectations

Tell students:

"When you come back, you'll tell us what you learned. You can say one thing about how your helper helps kids at the playground."

- Keep it simple: 1–2 sentences or a drawing they can show.

Step 5: Send a Note to Parents

- Include:
  - o Role options and their descriptions
  - o Instructions for finding information (read, talk, watch)
  - o Clear reminder about the dress-up or prop expectation

  - Day of the presentation

Possible Strategies for implementation:

1. Role Choice = Ownership

- Let students pick their own role so they feel excited and connected.
- Tip: Offer visuals or flashcards for non-readers.

2. Home Support Made Easy

- Provide optional links or suggestions for videos and books to explore (age-appropriate).
- Encourage family conversations as valid "research."

3. Costumes = Flexible & Fun

- Reassure families: costumes can be simple (e.g., a paper badge, a toy stethoscope, or a hat).
- Let kids bring drawings or handmade props instead.

4. Presentation Support

- Practice how to speak in front of the class:

"I'm a .... I help by..................."

- Use a sentence strip or cue card if needed.

Now I Can

Objective: Students will self-assess and reflect on their learning by reviewing what they accomplished during the inclusive playground project. Using a checklist, they will recognize their ability to ask questions, understand accessibility, generate inclusive ideas, and explain their design thinking. This activity supports the development of metacognition, confidence, and ownership of learning in a student-friendly, age-appropriate way.

Activity Instructions

Step 1: Create a Calm, Supportive Reflection Moment

- Begin with:

"Let's take a few minutes to think about everything you learned and how you became playground helpers!"

- Set a positive tone: soft music, seated in a circle, or quiet desk time.

Step 2: Introduce the "Now I Can" Checklist

- Show the checklist and read each item aloud slowly, one at a time.
- After each one, pause and ask:

"Can you give me a thumbs up if you can do this now?"

"Can someone share how they did this during our project?"

This supports oral reflection and includes students who may not write confidently yet.

Step 3: Reflect with Drawings or Sentences

- Pass out the checklist with space for them to check off or color stars next to what they did.
- Ask students to draw or write one thing they are proud of:

"What is something you made, said, or learned that helped others?"

This gives a visual and personal way to reflect.

Step 4: Partner Sharing

- Prompt students to turn to a partner and say:

"One thing I learned about helping all kids play is..."

"Next time, I would change..."

Develops speaking and listening skills.

Step 5: Celebrate Growth

- Say:

"Look at all you've done! You're real playground designers who think about everyone!"

- You may give a simple "I'm an Inclusive Engineer" badge or sticker.

Relation to Standards

1. **NGSS – Next Generation Science Standards**

K-2-ETS1-1

Standard: Ask questions, make observations, and gather information about a situation people want to change to define a simple problem.

Students reflect on their ability to ask questions and identify problems with existing playgrounds.

K-2-ETS1-2

Standard: Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function.

Reflection includes recalling their design drawings and explaining what they planned.

K-2-ETS1-3

Standard: Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.

Students think about what worked well and what they would improve in their playground models.

2. **Common Core – ELA Standards**

CCSS.ELA-LITERACY.SL.1.1

Standard: Participate in collaborative conversations with diverse partners about grade 1 topics and texts.

Through partner sharing and group discussions, students articulate their learning and listen to others.

CCSS.ELA-LITERACY.W.1.8

Standard: With guidance and support, recall information from experiences or gather information to answer a question.

Students recall and reflect on their project experience in both drawing and writing.

3. **ISTE – Technology Standards for Students**

1. Empowered Learner (1a, 1d)

Students articulate personal learning goals, reflect on the learning process, and demonstrate learning in multiple ways.

The checklist allows students to show what they learned and how they contributed, encouraging self-awarenessand ownership.

SEL & 21st Century Skills

- Self-awareness: Recognizing what they learned and what they can improve
- Communication: Expressing ideas clearly through speaking, drawing, or writing
- Empathy & Inclusion: Understanding how their design helped others